Week 05

UNIVERSITAS AMIKOM YOGYAKARTA

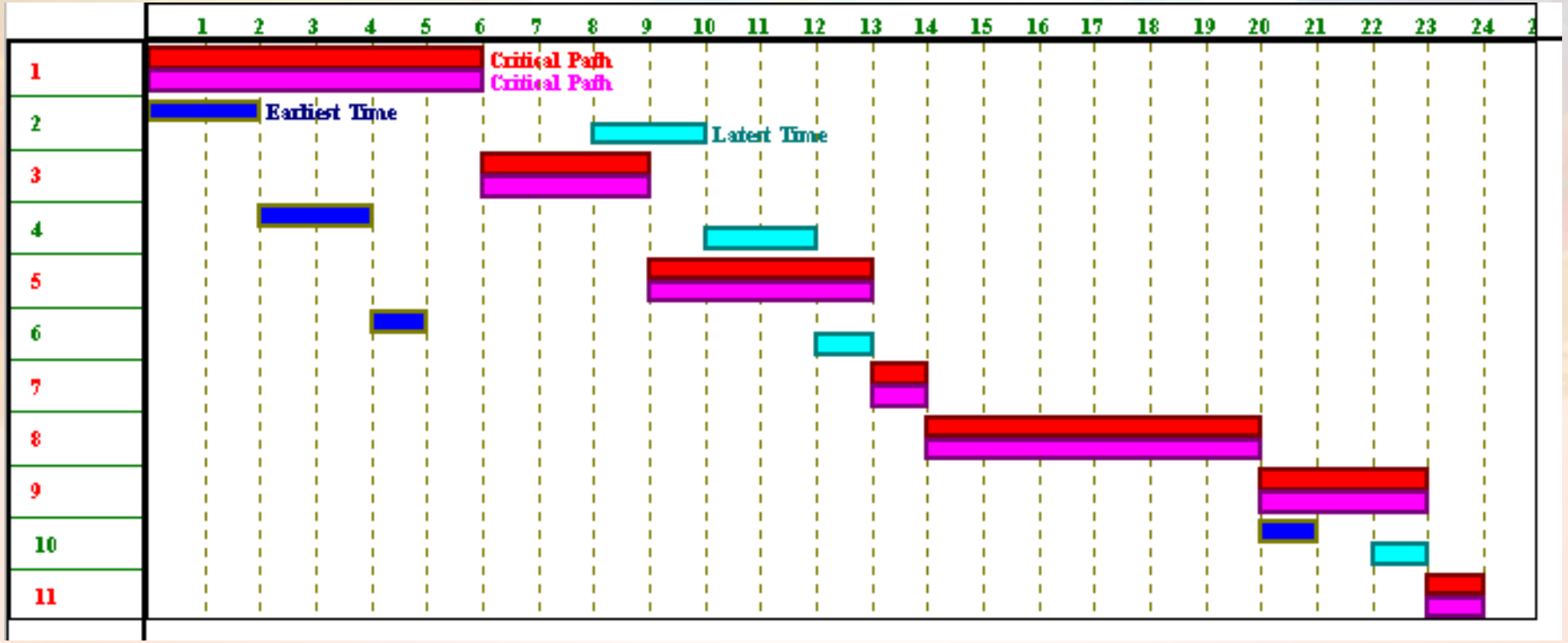
TIME & SCHEDULE MANAGEMENT

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1- Gantt Chart

- Graph or bar chart with a bar for each project activity that shows passage of time
- Provides visual display of project schedule



HISTORY OF CPM/PERT

- **Critical Path Method (CPM)**
  - E I Du Pont de Nemours & Co. (1957) for the construction of a new chemical plant and maintenance shutdown
  - Deterministic task times
  - Activity-on-node network construction
  - Repetitive nature of jobs
- **Project Evaluation and Review Technique (PERT)**
  - U S Navy (1958) for the POLARIS missile program
  - Multiple task time estimates (probabilistic nature)
  - Activity-on-arrow network construction
  - Non-repetitive jobs (R & D work)

PROJECT NETWORK

Network analysis is the general name given to certain specific techniques which can be used for the planning, management and control of projects

Use of nodes and arrows

❑ Arrows An arrow leads from tail to head directionally, indicating

ACTIVITY, a time-consuming effort that is required to perform a part of the work.

❑Nodes ● A node is represented by a circle. Indicate EVENT, a point in time where one or more activities start and/or finish.

Activity

- A task or a certain amount of work required in the project
- Requires time to complete
- Represented by an arrow

- **Dummy Activity**
  - Indicates only precedence relationships
  - Does not require any time or effort

Nodes



ES	ID Activity	EF
SL	resource	SL
LS	Duration (D)/ t	LF

PROJECT NETWORK

- **Event**
  - Signals the beginning or ending of an activity
  - Designates a point in time
  - Represented by a circle (node)
- Network
  - Shows the sequential relationships among activities using nodes and arrows

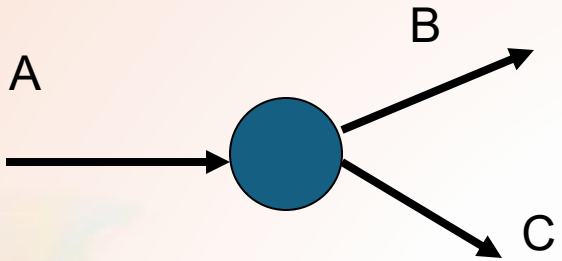
- **Activity-on-node (AON)**

  Nodes represent activities, and arrows show precedence relationships

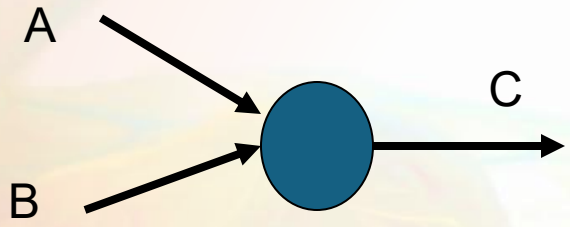
- **Activity-on-arrow (AOA)**

  arrows represent activities and nodes are events for points in time

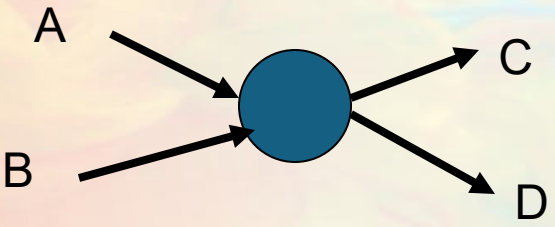
Situations in a network diagram



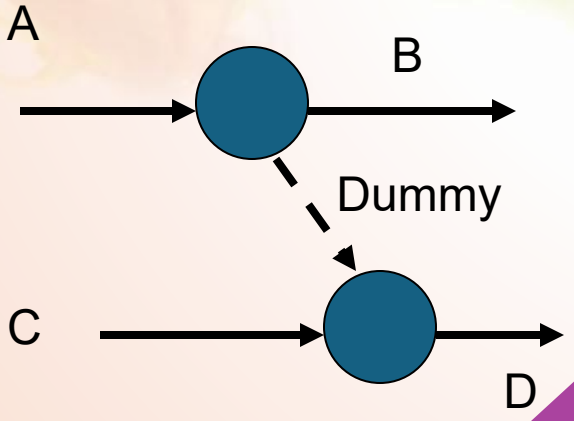
A must finish before either B or C can start



both A and B must finish before C can start



both A and C must finish before either of B or D can start



A must finish before B can start

both A and C must finish before D can start

NETWORK EXAMPLE

Illustration of network analysis of a minor redesign of a product and its associated packaging.

The key question is: How long will it take to complete this project ?

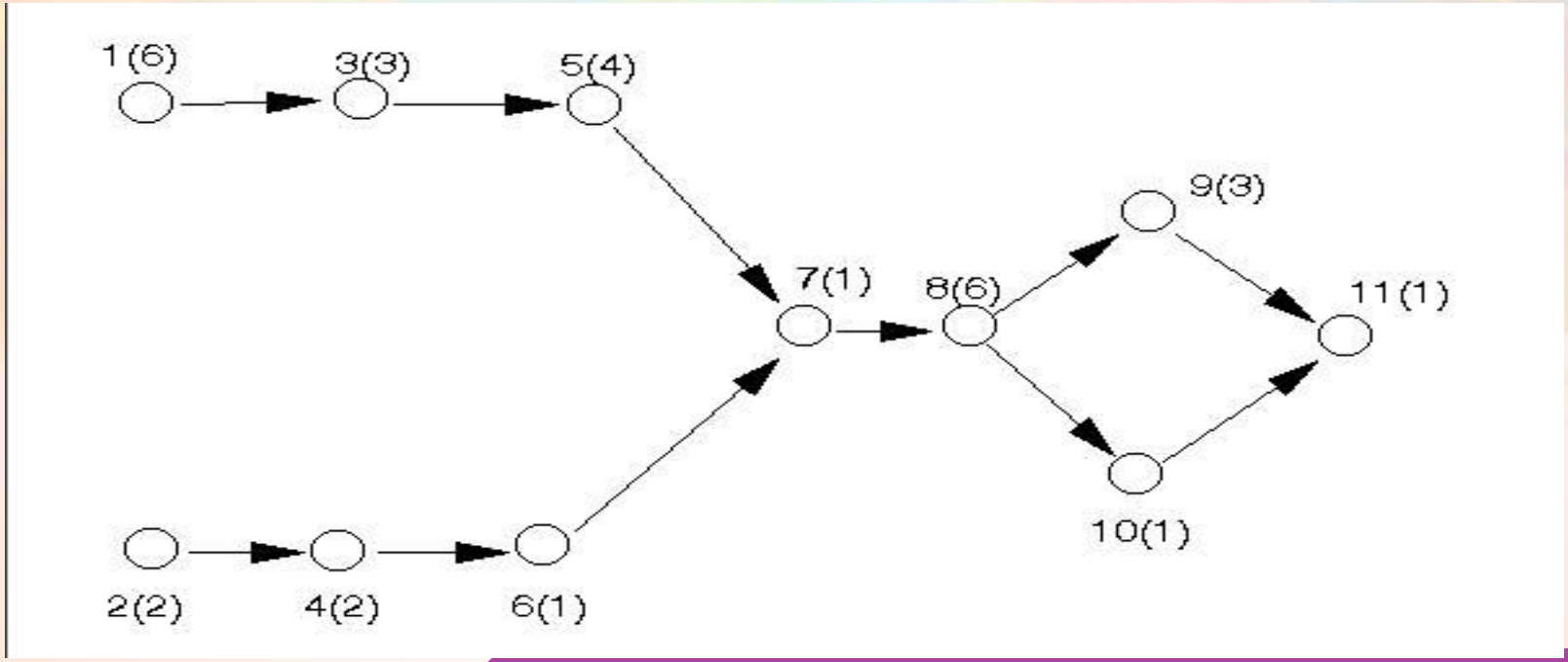
Activity number		Completion time (weeks)
1	Redesign product	6
2	Redesign packaging	2
3	Order and receive components for redesigned product	3
4	Order and receive material for redesigned packaging	2
5	Assemble products	4
6	Make up packaging	1
7	Package redesigned product	1
8	Test market redesigned product	6
9	Revise redesigned product	3
10	Revise redesigned packaging	1
11	Present results to the Board	1

For clarity, this list is kept to a minimum by specifying only immediate relationships, that is, relationships involving activities that "occur near to each other in time".

Activity number	must be finished before	Activity number	can start
1		3	
2		4	
3		5	
4		6	
5,6		7	
7		8	
8		9	
8		10	
9,10		11	

Questions to prepare activity network

- ☐ Is this a Start Activity?
- ☐ Is this a Finish Activity?
- ☐ What Activity Precedes this?
- ☐ What Activity Follows this?
- ☐ What Activity is Concurrent with this?



CPM calculation

- Path
  - A connected sequence of activities leading from the starting event to the ending event
- Critical Path
  - The longest path (time); determines the project duration
- Critical Activities
  - All of the activities that make up the critical path

Forward Pass

- **Calculation begin start to finish**
- Earliest Start Time (ES)
  - earliest time an activity can start
  - ES = maximum EF of immediate predecessors
- Earliest finish time (EF)
  - earliest time an activity can finish
  - earliest start time plus activity time

$$EF = ES + t$$

Backward Pass

- **Calculation begin finish to start**
- Latest Start Time (LS)

  Latest time an activity can start without delaying critical path time

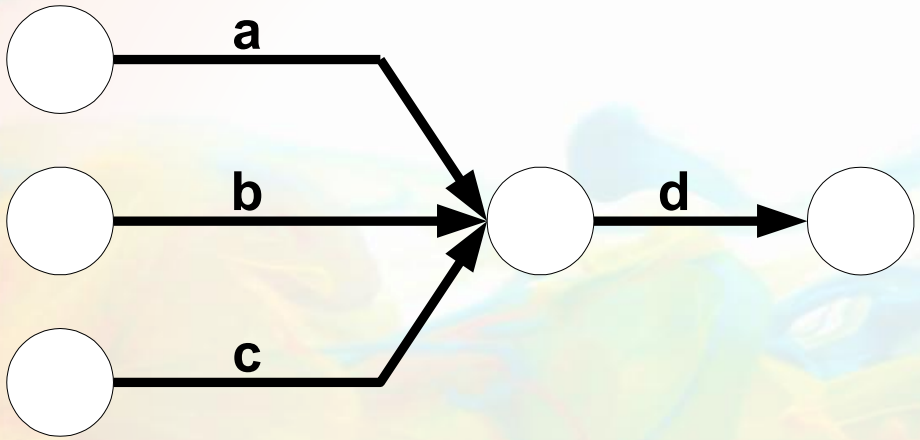
  $LS = LF - t$

- Latest finish time (LF)

  latest time an activity can be completed without delaying critical path time

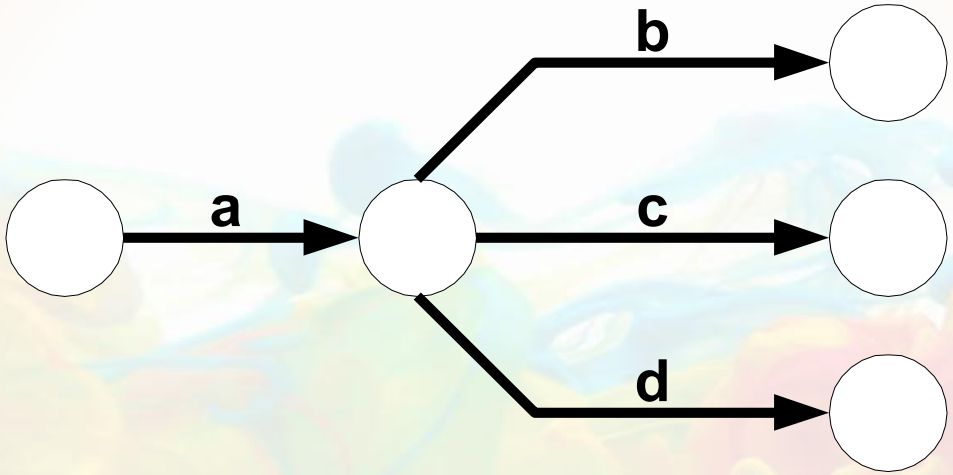
  LS = minimum LS of immediate predecessors

Sink Point



- If EF(c) > EF(b) > EF(a), then ES(d) = EF(c)

Burst Point



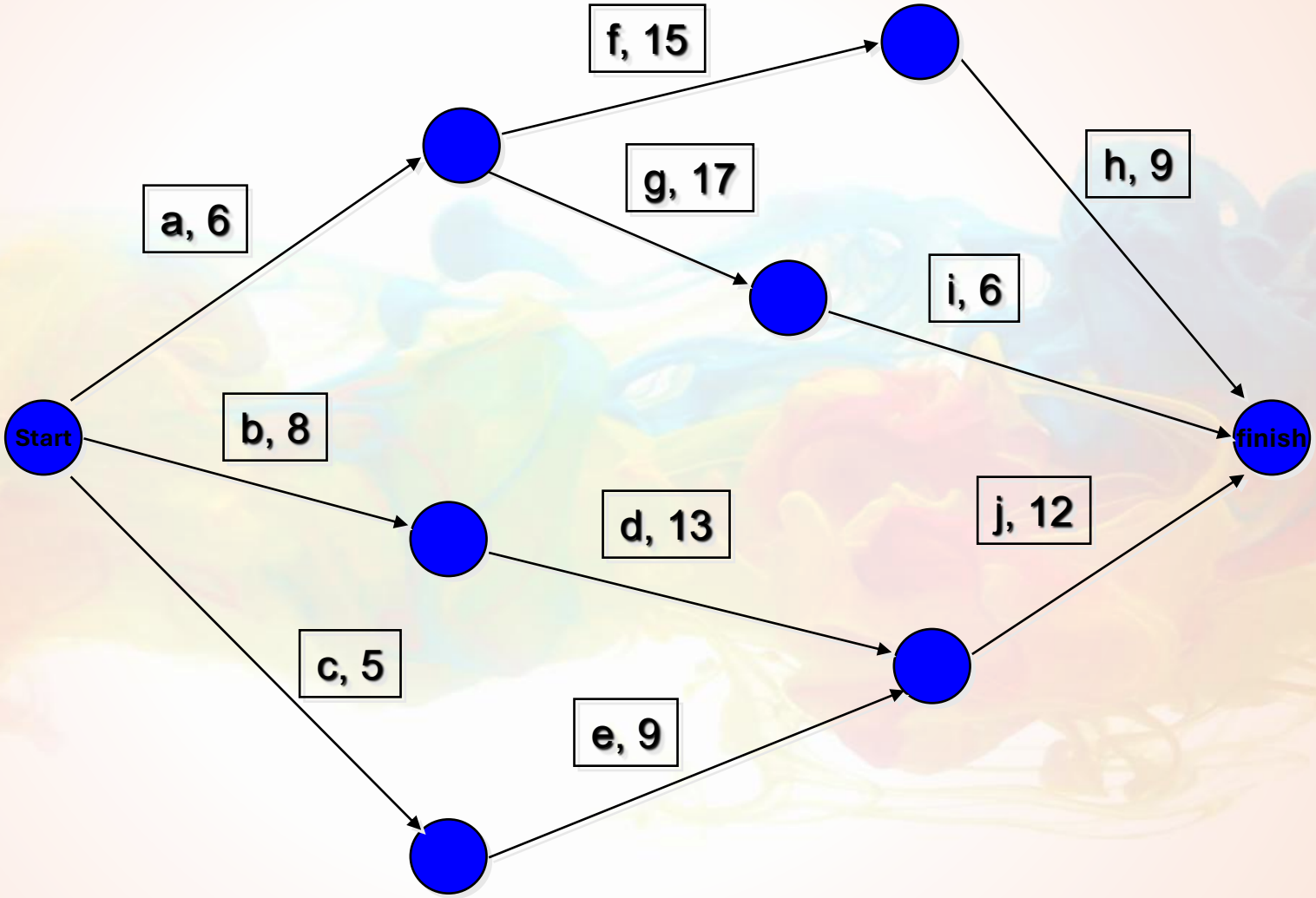
- If LS(b) < LS(c) < LS(d) Then LF(a) = LS(b)

CPM ANALYSIS

- Draw the CPM network
- Analyze the paths through the network
- Determine the float for each activity
  - Compute the activity's float

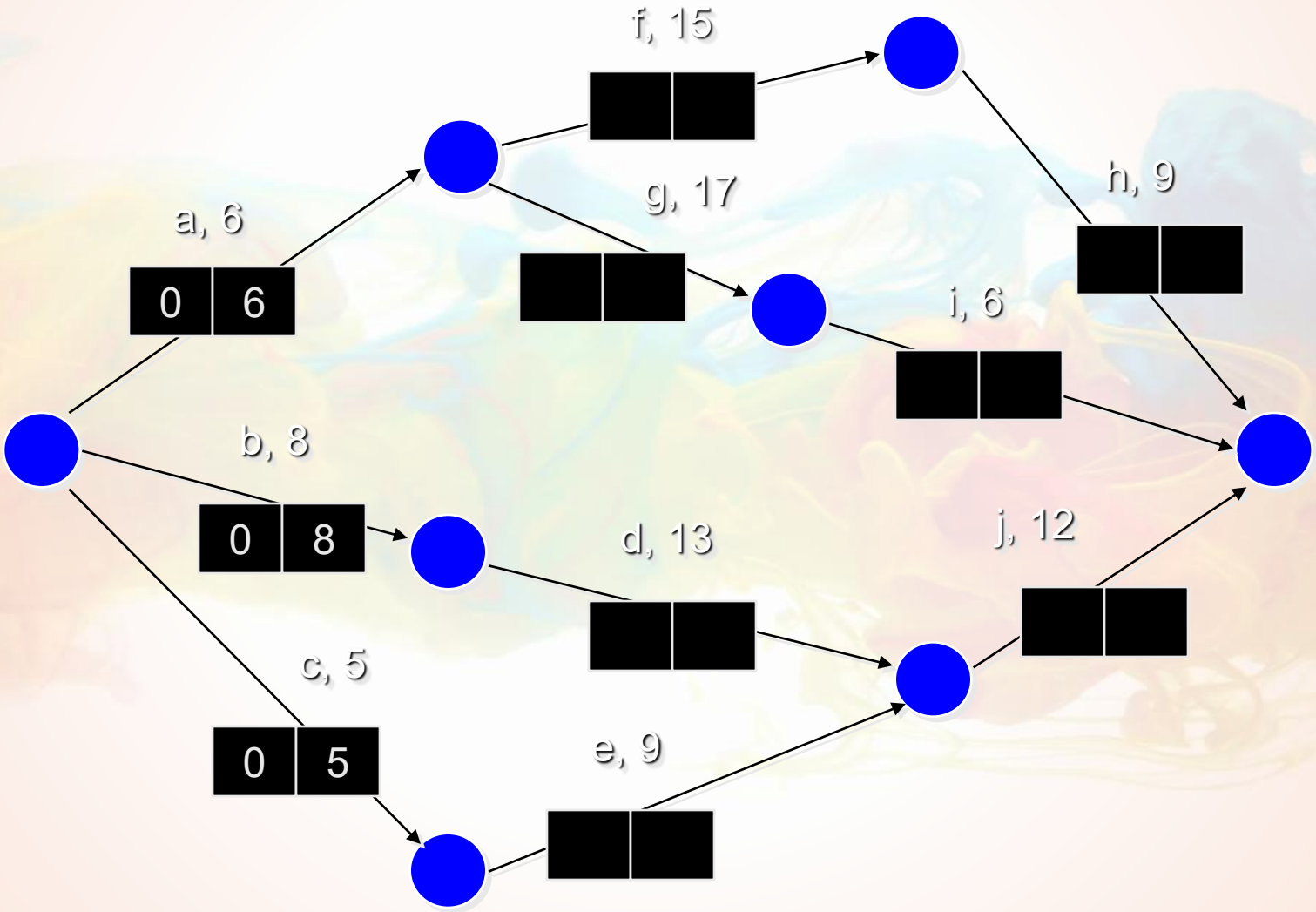
    float = LS - ES = LF - EF
  - Float is the maximum amount of time that this activity can be delay in its completion before it becomes a critical activity, i.e., delays completion of the project
- Find the critical path is that the sequence of activities and events where there is no "slack" i.e.. Zero slack
  - Longest path through a network
- Find the project duration is minimum project completion time

CPM EXAMPLE:



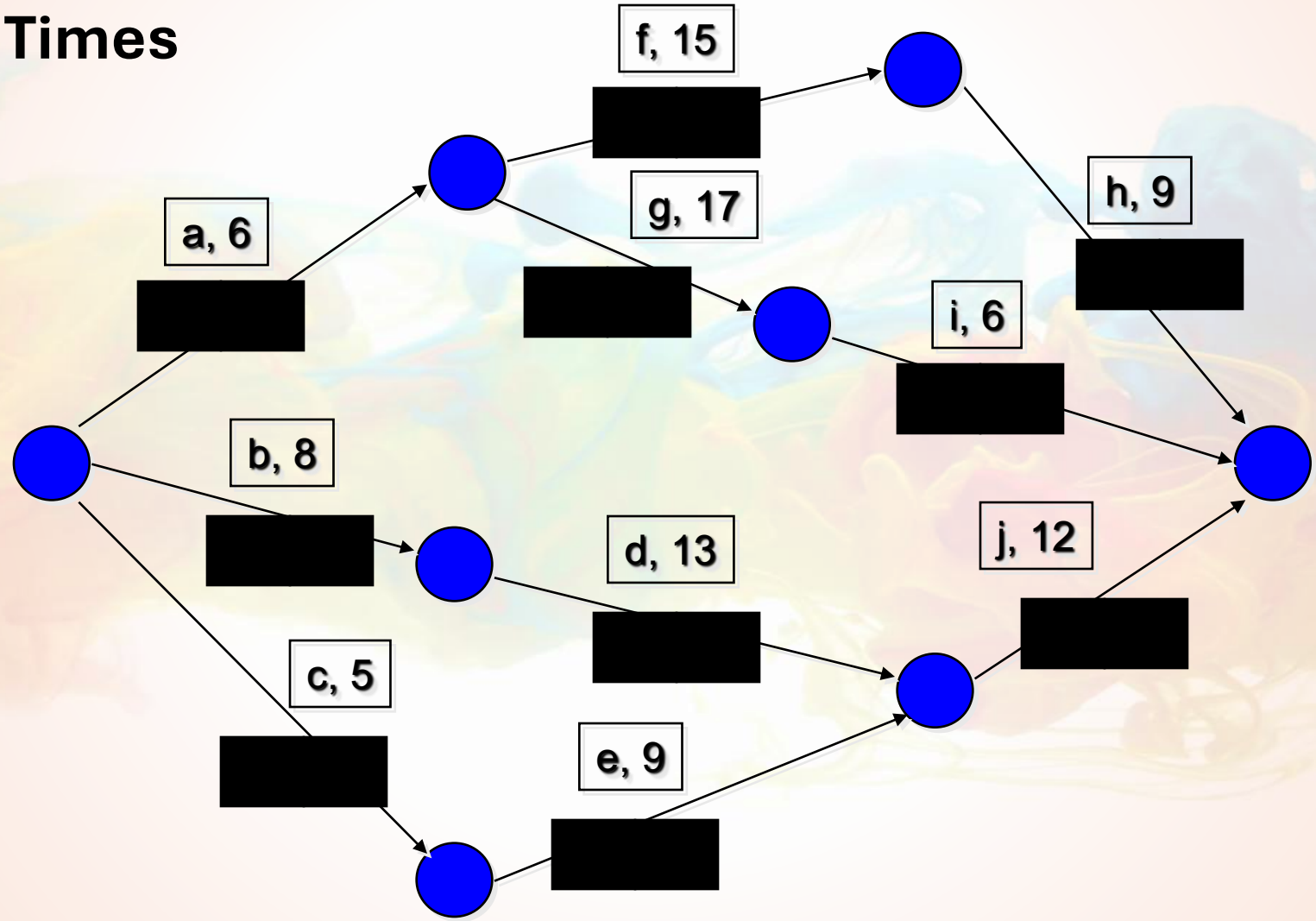
CPM Network

CPM Example ☐ES and EF Times



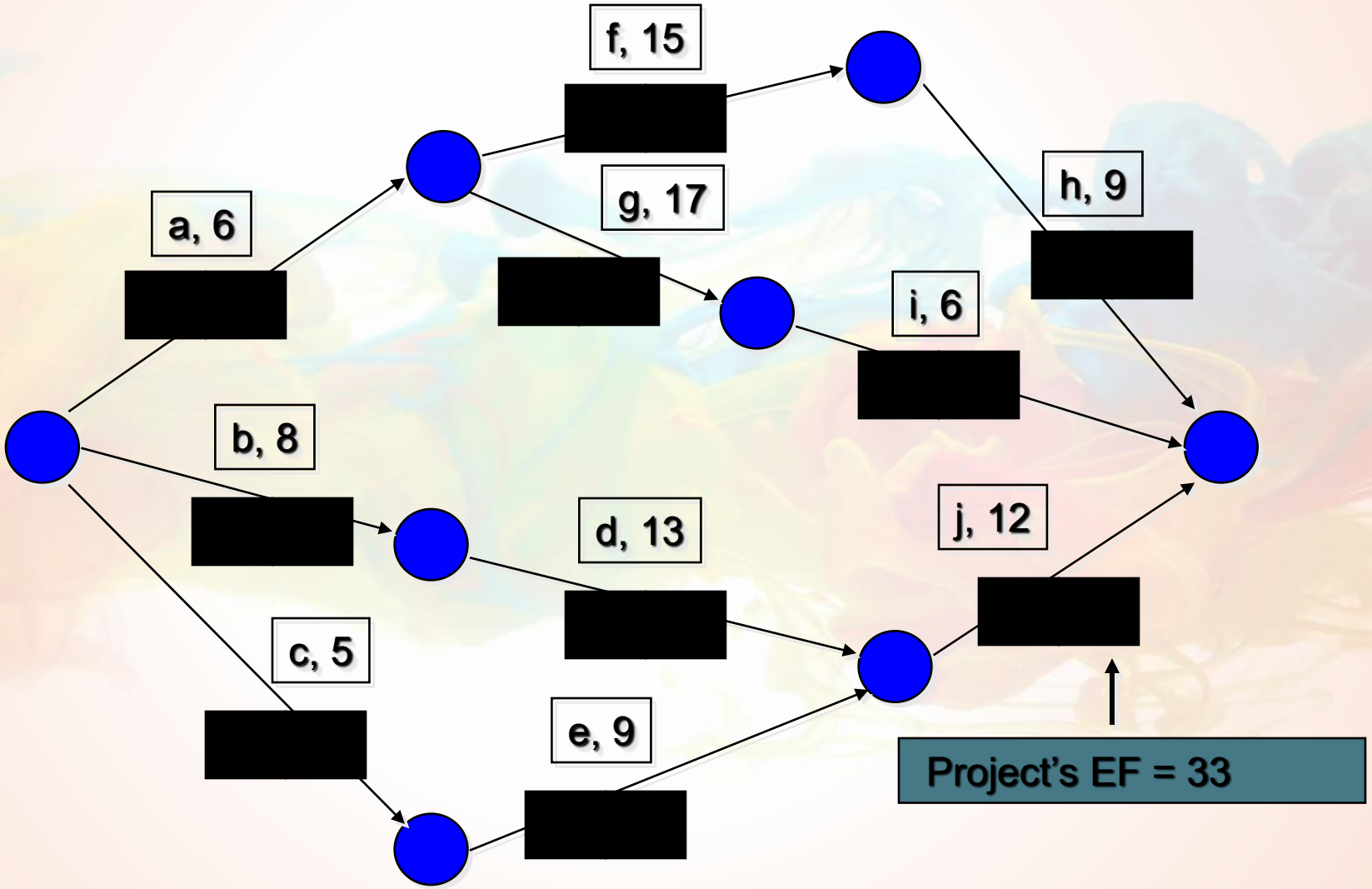
CPM EXAMPLE

ES and EF Times



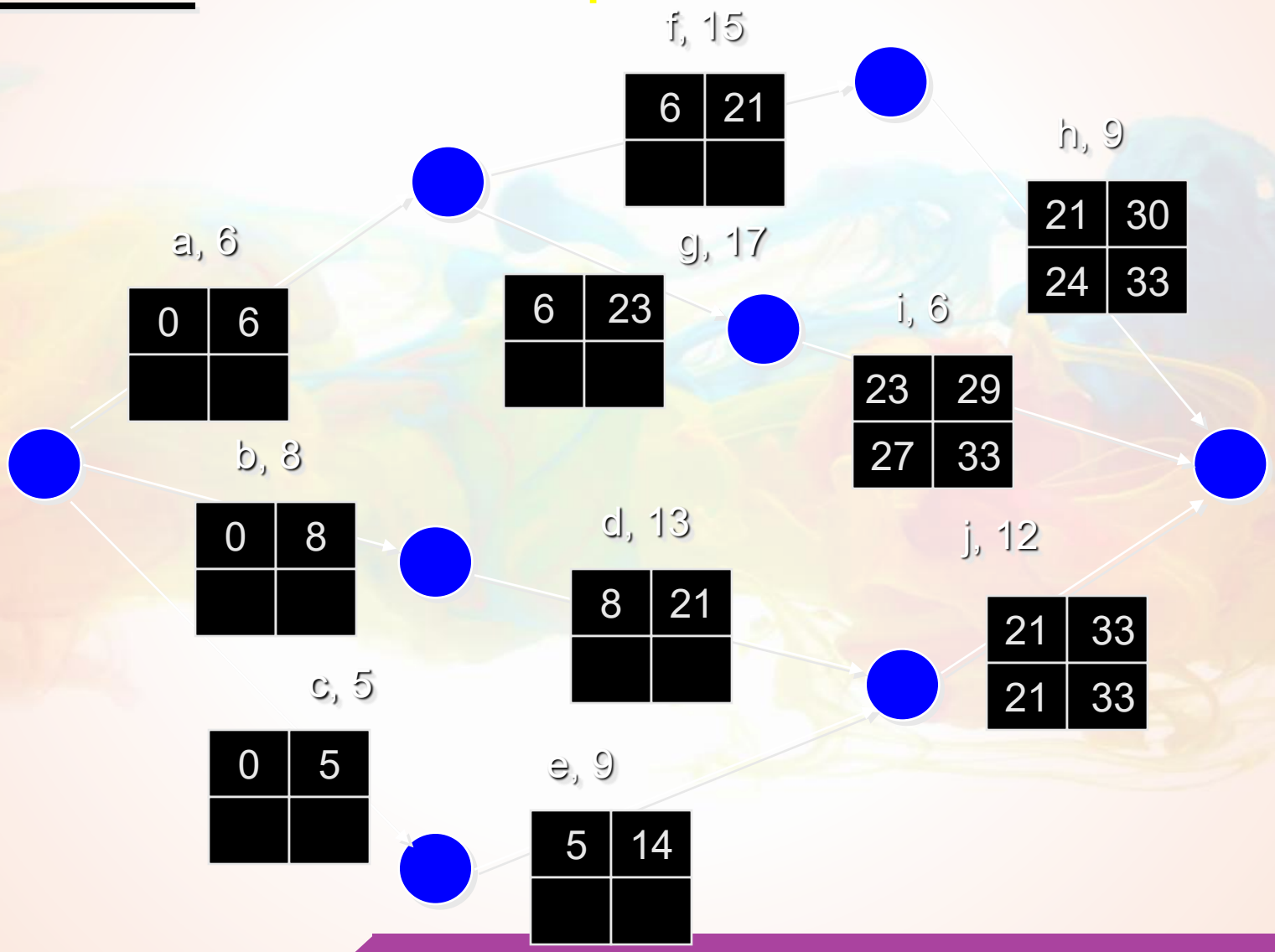
CPM Example

☐ES and EF Times

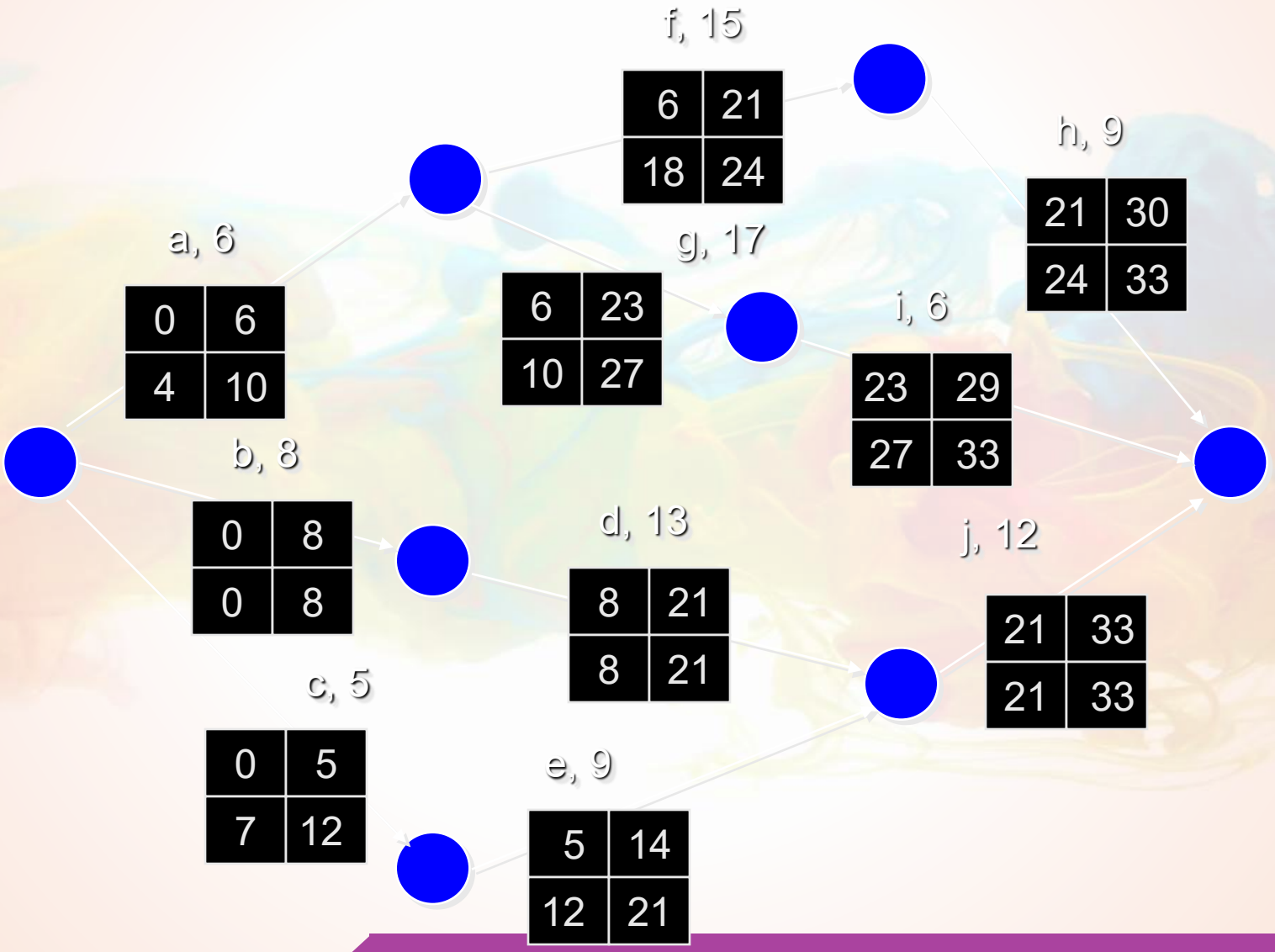


CPM Example

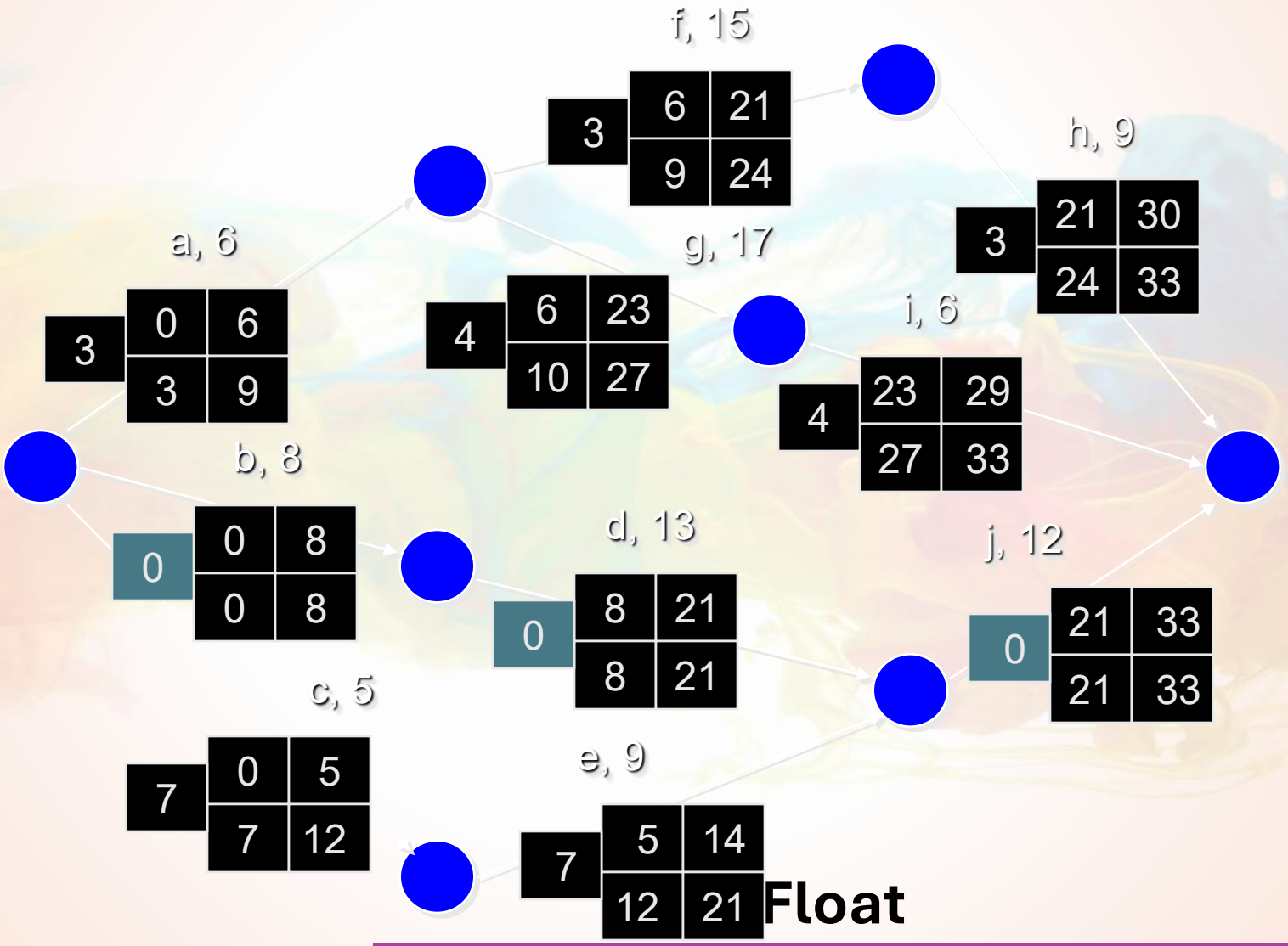
LS and LF Times



CPM Example LS and LF Times



CPM Example


f, 15
3
6 21
9 24
h, 9
3
21 30
24 33
a, 6
3
0 6
3 9
g, 17
4
6 23
10 27
i, 6
4
23 29
27 33
b, 8
0
0 8
0 8
d, 13
0
8 21
8 21
j, 12
0
21 33
21 33
c, 5
7
0 5
7 12
e, 9
7
5 14
12 21
Float

PERT ANALYSIS

- PERT is based on the assumption that an activity's duration follows a probability distribution instead of being a single value
- Three time estimates are required to compute the parameters of an activity's duration distribution:
  - pessimistic time (t_p) - the time the activity would take if things did not go well
  - most likely time (t_m) - the consensus best estimate of the activity's duration
  - optimistic time (t_o) - the time the activity would take if things did go well

Mean (expected time): $t_e = \frac{t_p + 4\,t_m + t_o}{6}$

Variance: $V_t = \sigma^2 = \left(\frac{t_p - t_o}{6}\right)^2$

Step of PERT analysis

- Draw the network.
- Analyze the paths through the network and find the critical path.
- The length of the critical path is the mean of the project duration probability distribution which is assumed to be normal
- The standard deviation of the project duration probability distribution is computed by adding the variances of the critical activities (all of the activities that make up the critical path) and taking the square root of that sum
- Probability computations can now be made using the normal distribution table.

Probability computation

Determine probability that project is completed within specified time

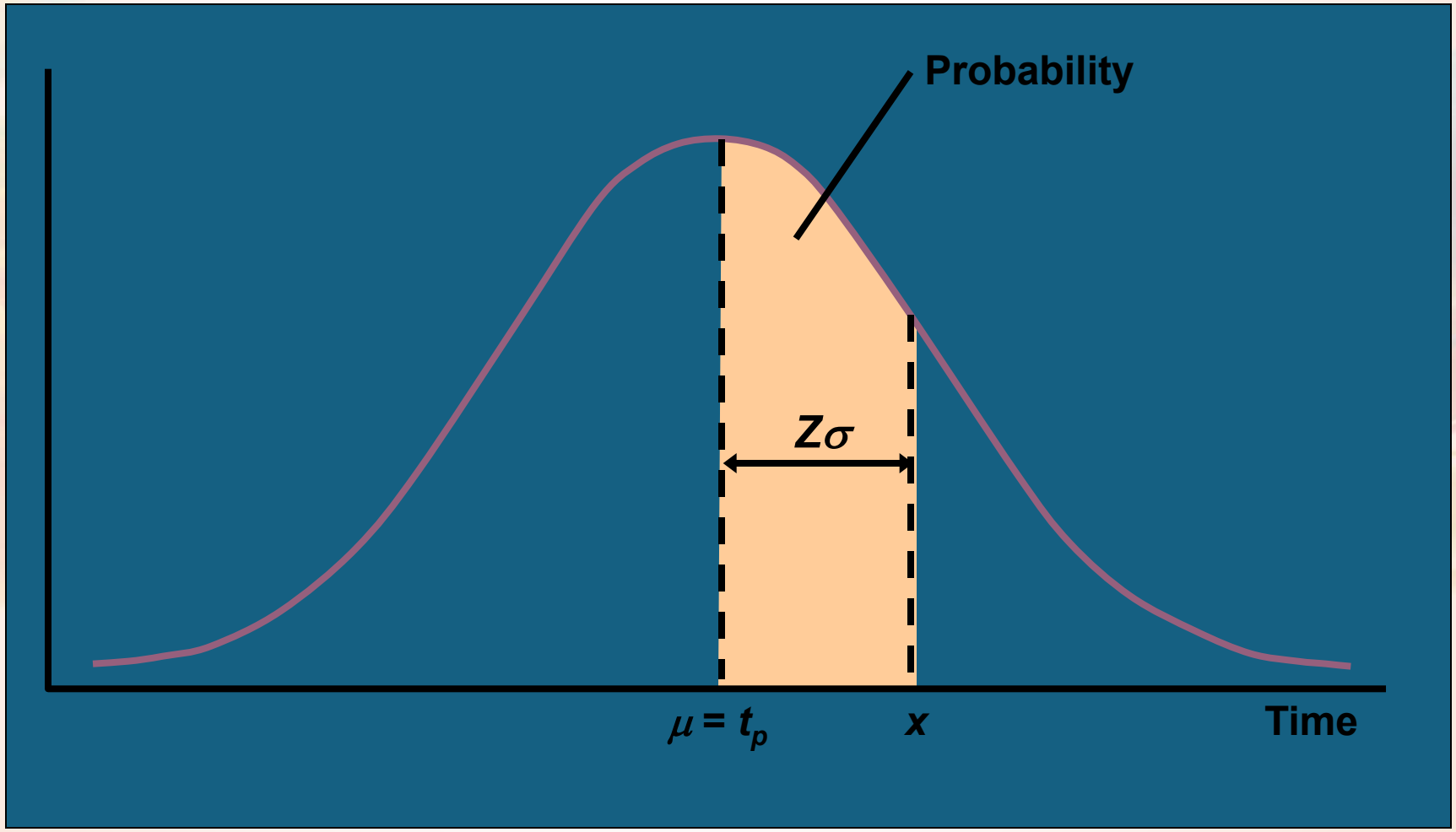
$$Z = \frac{x - \mu}{\sigma}$$

where $\mu = t_p$ = project mean time

σ = project standard mean time

x = (proposed) specified time

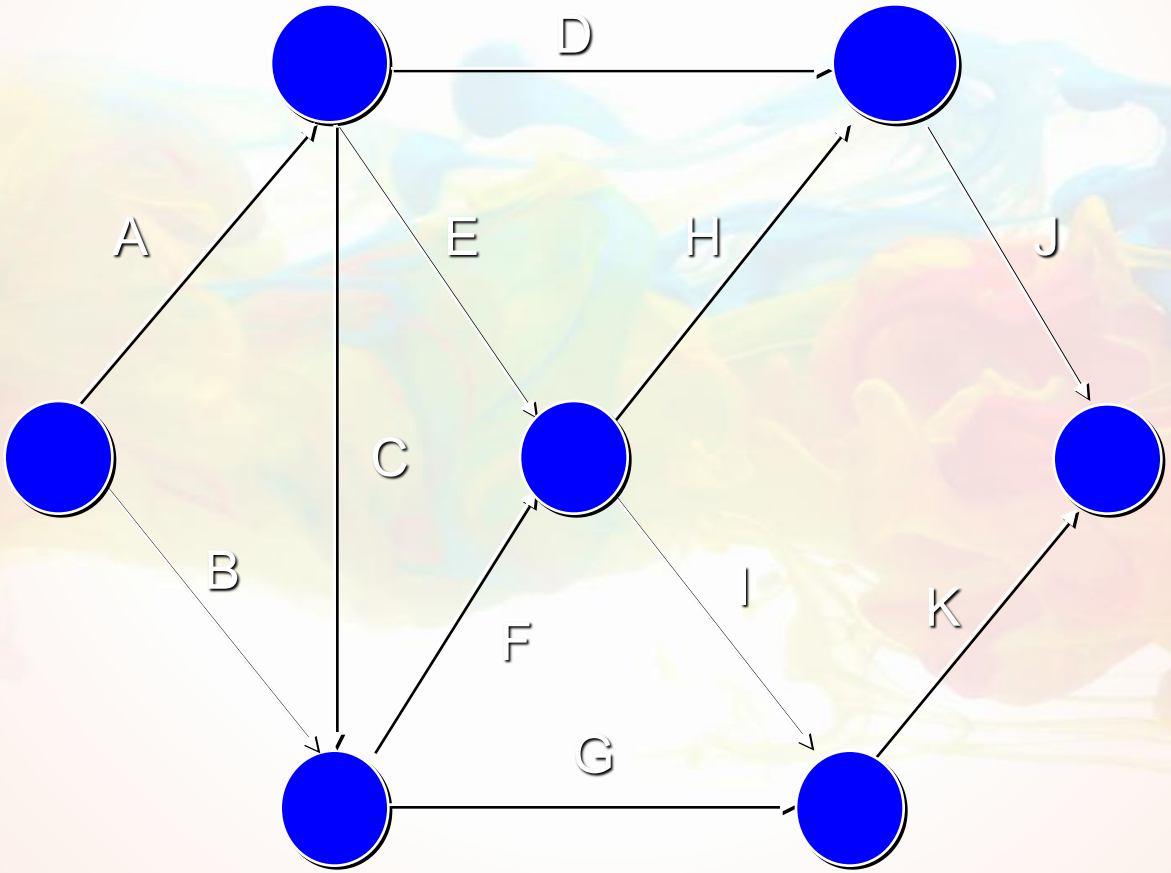
Normal Distribution of Project Time



PERT Example

Activity	Immed. Predec.	Optimistic Time (Hr.)	Most Likely Time (Hr.)	Pessimistic Time (Hr.)
A	--	4	6	8
B	--	1	4.5	5
C	A	3	3	3
D	A	4	5	6
E	A	0.5	1	1.5
F	B,C	3	4	5
G	B,C	1	1.5	5
H	E,F	5	6	7
I	E,F	2	5	8
J	D,H	2.5	2.75	4.5
K	G,I	3	5	7

PERT Example--PERT Network



PERT Example

Activity	Expected Time	Variance
A	6	4/9
B	4	4/9
C	3	0
D	5	1/9
E	1	1/36
F	4	1/9
G	2	4/9
H	6	1/9
I	5	1
J	3	1/9
K	5	4/9

Activity	ES	EF	LS	LF	Slack
A	0	6	0	6	0 *critical
B	0	4	5	9	5
C	6	9	6	9	0 *
D	6	11	15	20	9
E	6	7	12	13	6
F	9	13	9	13	0 *
G	9	11	16	18	7
H	13	19	14	20	1
I	13	18	13	18	0 *
J	19	22	20	23	1
K	18	23	18	23	0 *

PERT Example

$$V_{path} = V_A + V_C + V_F + V_I + V_K$$
$$= 4/9 + 0 + 1/9 + 1 + 4/9$$
$$= 2$$

$$\sigma_{path} = 1.414$$

$$z = (24 - 23)/\sigma = (24\text{-}23)/1.414 = .71$$

From the Standard Normal Distribution table:

$$P(z \leq .71) = .5 + .2612 = .7612$$